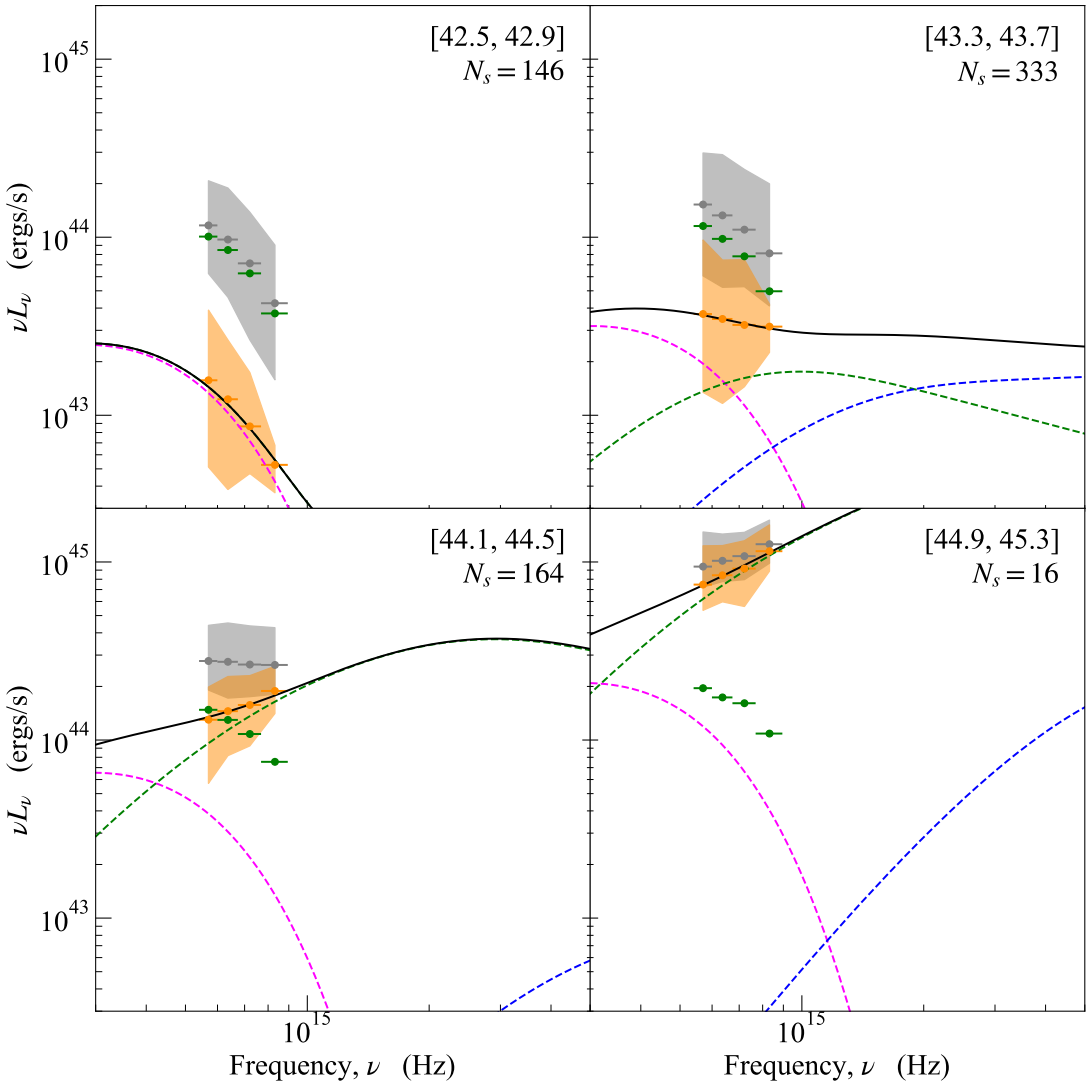

Figure A1. Zoom-in on the optical/UV region of the SEDs from Fig. 5, for select luminosity bins (as indicated in the top right corners). Here we show the stacked data for the non-decomposed HSC fluxes (i.e host + AGN) in grey, with the decomposed stacks (AGN only) in orange, highlighting the necessity of the HSC decomposition. The green points show the galaxy only stacks, with the error region left off for clarity, to highlight that the evolution in the SEDs is driven by the AGN rather than the host. In all, apart from the highest (likely super-Eddington), luminosity bins the inferred power and optical SED shape would be incorrect with no decomposition.

the stacks are not obscured, ruling out $N_H \gtrsim 3 \times 10^{21}$ cm^{-2} at 99 % confidence.

However, the above is assuming all sources have some intrinsic obscuration. Hence, we now also test the case where only some of the sources are obscured, contaminating our stacks. Defining the XSPEC model PCF*POW, we set a covering fraction of 0.5, effectively assuming 50 % of the sources within the stacks are obscured. The results for this model are shown as the dotted coloured lines in Fig. B1. In this case our stacks are clearly unable to constrain the column-density, with the results being consistent with both completely unobscured and half of the sources fully obscured.

To test whether a set of heavily absorbed sources within the stacks could feasibly turn a complex spectrum into what appears as a single power-law, we perform a test using the SED model from luminosity bin $\nu L_{3500} \in [44.1, 44.5]$ from Fig. 5. Our choice falls on this bin as it give a relatively typical AGN SED, with a clear contribution from the soft X-ray excess, complicating the shape beyond a single power-law. Taking this SED model, we now apply the partial covering fraction absorption model PCF, setting the covering fraction to 0.5 and column-density $N_H = 10^{24}$ cm^{-2} (i.e assuming half the sources within the stacks are heavily obscured). These results are shown in Fig. B2 where we can clearly see that, although the overall normalisation is altered, the spectral shape would not turn to a single power-law if half the objects within the stack are obscured. Certainly, the soft X-ray excess would still be observable within the stacks, leading us to conclude that the single power-law shape in our low-luminosity stacks must be intrinsic to the AGN themselves; not an artefact from spurious obscuration.

APPENDIX C: ALTERNATE MASS BINS

Here we present the stacked SEDs for the mass bin below ($\log M/M_\odot \in [7.5, 8.0]$ - Fig. C1 and above ($\log M/M_\odot \in [8.5, 9.0]$ - Fig. C2) our fiducial mass bin, confirming that the same change in accretion state is seen beyond the single bin.

We note that the luminosity binning differs between the mass bins, as we optimise this to give identical bin widths while also

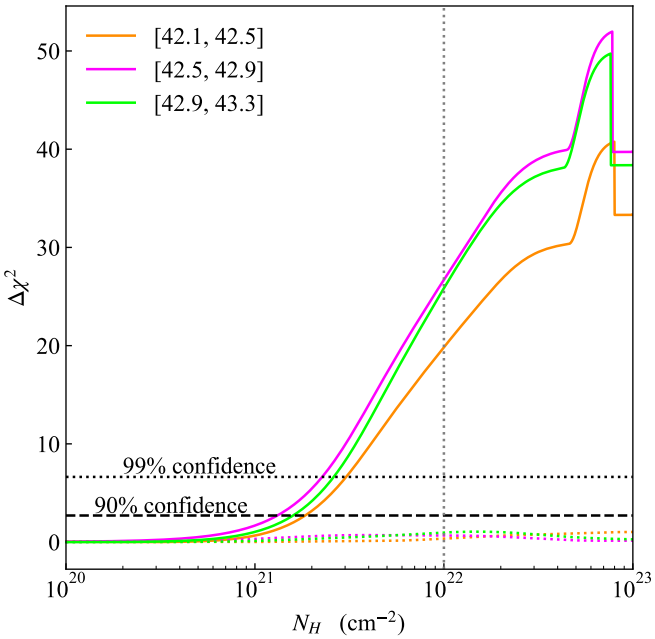

Figure B1. $\Delta\chi^2$ curves for an absorbed power-law model, using the X-ray stacks for the three lowest luminosity bins in Fig. 5. The solid coloured lines show the case where the absorber covers all sources within the stacks (i.e PHABS*POW), while the dotted coloured lines show the case where only half the sources are obscured (i.e PCF*POW). The horizontal dashed line and dotted line show the 90 % and 99 % confidence levels respectively, while the vertical dotted line highlights a column-density of $N_H = 10^{22}\,\mathrm{cm}^{-2}$. It is clear that we completely rule out the X-ray stacks being obscured on average, however in the case where only half the sources are obscured we can make no upper constraints on N_H.

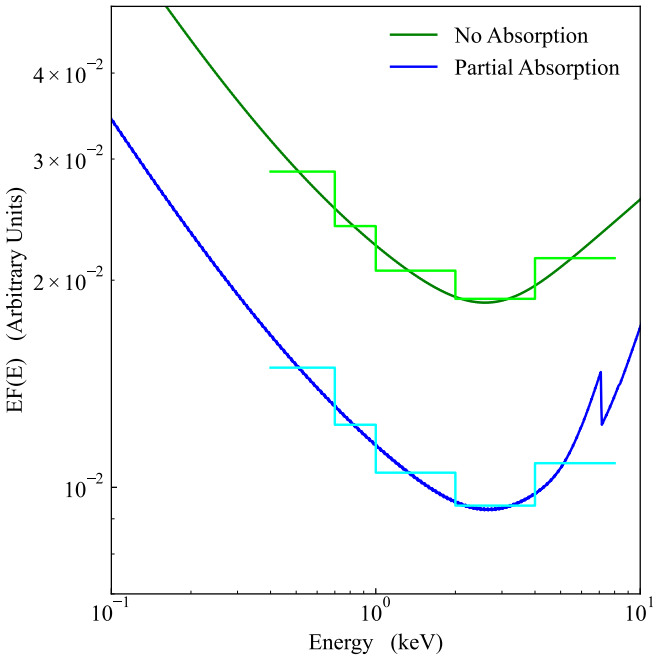

Figure B2. The X-ray portion of the model SED from luminosity bin $\nu L_{3500} \in [44.1, 44.5]$ from Fig. 5. The solid green line shows the unabsorbed model, while the solid blue line shows the same model after including a 0.5 covering fraction absorber at $N_H = 10^{24}\,\mathrm{cm}^{-2}$ in order to asses the impact of half the objects in a stack being heavily obscured. The stepped lines show the models on the energy grid of the eROSITA stacks.

avoiding an uneven luminosity distribution in the edge bins. As with the fiducial mass bin, the stacks are too coarse to give well defined SED parameters. Nonetheless, the same trend is clear, with a receding disc as $\dot{m}$ reduces, eventually collapsing into a flow dominated by the X-ray plasma at low $\dot{m}$. The SED fit showing this are given in table C1.

This paper has been typeset from a TEX/LATEX file prepared by the author.

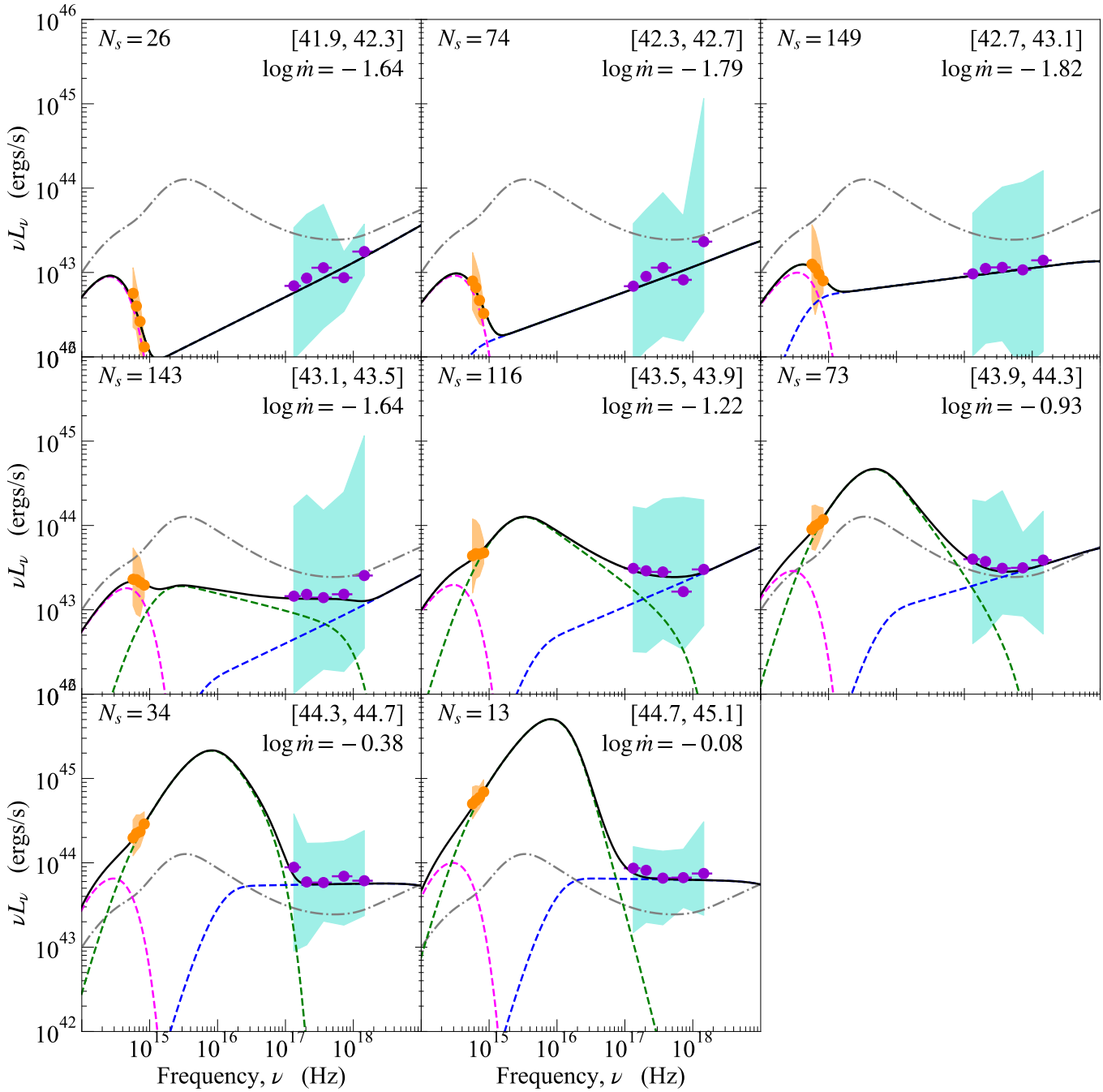

Figure C1. Same as Fig. 5, but for the mass bin $\log M/M_\odot \in [7.5, 8.0]$